

CHAPTER 6: GEOMETRICAL OPTICS

1. An upright object is placed in front of a spherical mirror and the magnification of the image formed is 3. If the image distance is 40 cm from the center of curvature of the mirror and an upright image is formed,
 - a) state the type of spherical mirror.
 - b) sketch the formation of the image.

2. An object is placed at 10 cm from a concave mirror with a radius of curvature of 30 cm.
 - a) Calculate the position of image for that object.
 - b) State the characteristics of that image

[-30 cm]

3. An object is placed at a distance of 30 cm in front of a curve mirror. If a virtual image is formed at a distance of 5 cm from the mirror, calculate the value of its radius of curvature and state the type of mirror.

[-12cm]

4. A spherical mirror produces a virtual image with a linear magnification of 2. If the object is placed 10 cm from the mirror,
 - a) calculate the focal length of mirror.
 - b) identify the type of mirror.

[20 cm]

5. A concave mirror has a radius of curvature 30 cm. An object height 12 cm forms an enlarged image 20 cm from the mirror.
 - (i) Calculate the object distance.
 - (ii) Calculate the height of the image.
 - (iii) State the characteristics of the image.

[8.57 cm, 28 cm]

6. An object is placed 12 cm in front of a mirror. The real image formed is the same size as the object.
 - a) Determine the focal length of the mirror.
 - b) State the type of mirror.

[6 cm]

7. Sketch the image formation of a convex lens for an object that produces an upright image and magnified.

8. An insect of 1.5 cm tall is at 13.0 cm from a converging lens with a focal length of 13.5 cm. Determine the position, characteristics, and the height of the image produced.

[-351 cm, 40.5 cm]

9. An object of height 10 cm is placed 30 cm from a converging lens. The height of the virtual image formed is 20 cm. Calculate
 - (a) the image distance.
 - (b) the focal length of the lens.

[-60 cm, 60 cm]

10. A thin converging lens with refractive index 1.5 has both radii of curvature 20 cm. An object is placed 10 cm from the lens. Calculate the image distance.

[-20 cm]